Week 10 Final Project – Phase 1 (Planning and Design)

Project Scope: Inventory & Sales Management System

System Description

Our **Inventory & Sales Management System** is an integrated database solution that manages the end-to-end lifecycle of retail products within a warehouse or retail environment. It helps businesses efficiently monitors stock levels from procurement to point-of-sale, ensuring that every sale automatically updates inventory counts. The system provides a unified platform for tracking supplier deliveries, managing customer information, generating sales reports, and is capable of handling large datasets (100,000+ records), making it suitable for real-world business operations.

Key Functionality

- **Inventory & Supplier Tracking:** Manages stock levels, reorder points, and vendor contact details.
- **Sales Management:** Processes customer transactions and generates digital receipts.
- **Automated Stock Updates:** Deducts items from the Products table inventory immediately upon a confirmed sale.
- **Customer Relationship Management (CRM):** Maintains a database of customer contact info and purchase history.

Stakeholders

- **Project Manager:** Responsible for overseeing the database development lifecycle, ensuring team members meet deadlines, and finalizing all project documentation.
- **Warehouse/Inventory Manager:** The primary user of the inventory module. They are responsible for monitoring stock levels, identifying low-stock items, and coordinating with suppliers for replenishments.
- **Sales Associates/Cashiers:** Front-end users who interact with the sales module to process customer transactions and update the inventory in real-time.
- **System Developers (The Group):** The technical team (DBA, SQL Developer, etc.) responsible for the design, normalization, and maintenance of the relational database.
- **Suppliers:** External partners who provide the products. Their data is stored to facilitate procurement and track unit costs.
- **Customers:** The end-users whose purchase behavior generates the transactional data. Their information is captured for sales history and CRM purposes.

Core Data Processes

- **Sales-Inventory Linkage:** When a sale is finalized, the system creates a record in Sales_Items and triggers a reduction in the current_stock of the Products table.
- **Historical Pricing:** Stores the unit_price_at_sale to ensure that financial reports remain accurate even if product prices are updated later.

Members:

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